Unit – 13 Organic Nitrogen compounds

1. Option (a)
2. Option (b)
3. Option (a) only primary amides undergo hoffmann bromamide reaction
4. Option (d) both are wrong
5. $CH_3CH_3Br \xrightarrow{aq\ NaOH} CH_3CH_2\text{-}OH \xrightarrow{KMnO_4} CH_3\text{-}COOH \xrightarrow{NH_3} CH_3CONH_2 \xrightarrow{Br_2/NaOH} CH_3\text{-}NH_2$

   Option (c)
6. Option (c) 3° nitroalkane
7. Option (c)
8. Option (c) Suhiff's base
9. Option (b) p – nitrosation takes places, the product is $(CH_3)_2N$–C$_6$H$_4$–NO

10. $C_6H_5NH_2 + CH_3COOCOCH_3 \longrightarrow C_6H_5NHCOCH_3$

    acetanilide

    Option (d)
11. Option (d)
12. Option (a)
13. Option (a)

    $C_6H_5NO_2 \xrightarrow{Fe/HCl} C_6H_5NH_2 \xrightarrow{NaNO_2\ /\ HCl} C_6H_5\text{-}N_2Cl \xrightarrow{H_2O} C_6H_5\text{-}OH + N_2\uparrow + HCl$
14. (d)

    $C_6H_5NO_2 \xrightarrow{HNO_3\ /\ H_2SO_4} C_6H_4(NO_2)_2$

    1,3 - dinitrobenzene
15. (b)

    $CH_3CH_2CH_2C(CH_3)(H)NH_2 \xrightarrow{HONO} CH_3CH_2CH_2C(CH_3)(H)OH$
16. Option (b) blue solution

Answers.indd 318 11-01-2021 17:36:54